

2024 Engineering Salary Benchmark Report

Software Engineer Salaries in the United States — by role, level, and city.

Overview

Software engineers remain among the highest paid roles in the tech industry. Although salaries have dropped by 9% to 15% in some segments due to mass layoffs, hiring freezes, the acceptance of remote work, and the increasing use of AI to automate engineering tasks, the long-term trend is positive. According to the U.S. Bureau of Labor Statistics, the average computer programmer annual salary increased from \$61,540 in 2000 to \$91,610 in 2020.

Skills and experience in emerging technologies like AI, Machine Learning, Python, Kubernetes, and Docker are in extremely high demand. As a result, specialised engineers can command very high salaries, sometimes exceeding \$300,000 per year, especially when combined with strong systems-engineering skills (Linux, AWS, Kafka).

Salaries by Engineering Role

Role	Median Salary (USD/yr)	Key Skills
Software Engineer	\$166,000	Python, Java, SQL, Git, Linux
Security Engineer	\$139,050	Linux, AWS, Python, Bash
Hardware Engineer	\$133,271	C++, Verilog, Linux
DevOps Engineer	\$148,190	Kubernetes, Docker, AWS, CI/CD
Data Engineer	\$150,000	Python, SQL, Spark, Airflow
Machine Learning Engineer	\$177,238	Python, TensorFlow, Scikit-learn, MLOps
Cloud Architect	\$185,500	AWS, Azure, GCP, Terraform
Backend Engineer	\$162,000	Java, Go, Spring, Microservices

Table 1. Median U.S. software-engineering salaries by role, 2024.

Salaries by Career Level

Compensation scales steeply with seniority. Each company structures levels differently, but the Pave dataset reflects a consistent progression from Entry to Director.

Level	Track	Median Salary (USD/yr)
P1 (Entry)	Professional	\$115,416
P2 (Mid)	Professional	\$146,062
P3 (Mid)	Professional	\$166,000
P4 (Senior)	Professional	\$187,500

P5 (Senior)	Professional	\$219,350
P6 (Senior)	Professional	\$250,000
M3 (Manager)	Management	\$206,078
M4 (Manager)	Management	\$226,344
M5 (Director)	Management	\$240,000
M6 (Director)	Management	\$293,500

Table 2. Median compensation by Pave career level.

Salaries by U.S. Metro Area (Mid-level)

Geographic differences are significant — the gap between the most expensive and least expensive U.S. cities has shrunk by two-thirds between 2019 and 2021, but a clear premium remains for established tech hubs.

City	Salary	City	Salary
New York City, NY	\$189,000	Boston, MA	\$166,500
Seattle, WA	\$187,000	Houston, TX	\$165,000
San Francisco Bay, CA	\$184,500	St. Louis, MO	\$165,000
Albany, NY	\$180,000	Atlanta, GA	\$160,872
Baltimore, MD	\$175,050	Austin, TX	\$160,000
Sacramento, CA	\$175,000	Charlotte, NC	\$159,840
Portland, OR	\$169,832	Nashville, TN	\$159,300
Birmingham, AL	\$168,000	Denver, CO	\$159,200
Pittsburgh, PA	\$167,000	Dallas/Fort Worth, TX	\$157,000
Cincinnati, OH	\$156,000	Omaha, NE	\$130,000

Table 3. Median P3 (mid-level) software engineer salary by metro area.

AI and Machine Learning Compensation

AI-related job postings increased by 15.7% in the past six months as of January 2024, while software engineer and developer listings decreased by 2.9%. OpenAI publishes typical compensation for a level-five engineer with 10+ years of experience of approximately \$925,000 USD per year — a \$300,000 base salary plus \$625,000 in stock-based compensation. Some higher-level engineers earn as much as \$1,400,000. The lowest reported salary at OpenAI was \$210,000 for a software engineer with two to four years of experience.

Across the broader market, senior- and staff-level machine learning engineers have a base salary that is 7% to 8% higher than that of a software engineer of the same level. The gap shows up in equity rather than salary — equity is the real differentiator for technical talent in AI roles.

Methodology and Sources

Data is drawn from Pave (compensation management platform, ~8,000 customers) and Karat (technical interviewing, ~350,000 interviews globally). Skill frequencies were extracted from Indeed, Stack Overflow, and LinkedIn job postings during Q1 2024. All salaries are in USD per year unless noted. Technologies referenced include Python, Java, JavaScript, TypeScript, SQL, AWS, Azure, GCP, Kubernetes, Docker, Linux, Bash, Spark, Airflow, Kafka, Scikit-learn, TensorFlow, PyTorch, Git, CI/CD, Terraform, Microservices.